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Review Article

Research and prospect of ground-penetrating radar on the Moon

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Abstract

Ground-penetrating radar (GPR) has become a pivotal tool in lunar geological exploration, providing direct insight into the Moon's subsurface structure. This review provides a focused summary of GPR applications in the Apollo 17, Kaguya, and Chang'e lunar missions, synthesizing their key scientific findings and technological advancements. In terms of scientific discoveries, GPR observations have revealed a lunar subsurface far more complex than previously thought, characterized by a multi-layered stratigraphy of volcanic deposits, impact ejecta, and ancient buried regolith. Furthermore, GPR has identified numerous unique subsurface features, including lava tubes, buried impact craters, and bifurcated structures. Notably, radar data reveal that the lunar farside's regolith exhibits significantly lower attenuation of radio waves than the nearside's, due to its lower iron and titanium oxide content, which allows for greater penetration depths. The paper also highlights recent advances in lunar GPR data processing, including forward modeling, improved dielectric property estimation through methods such as hyperbolic fitting and direct comparison with returned samples, and the application of advanced migration and machine learning-based imaging methodologies. These techniques have been crucial for converting radar travel times to depth, enhancing image resolution, and accurately characterizing lunar materials. Different GPR systems offer a trade-off between penetration depth and resolution. Orbital sounders can see deep into the subsurface with low resolution, whereas surface rovers provide high-resolution data limited to shallow depths. Therefore, a comprehensive understanding of the lunar subsurface requires a multi-frequency, data-fusion approach. Looking ahead, GPR will be indispensable for future resource-prospecting missions, such as mapping water ice deposits, and for the selection of sites for lunar bases.

Keywords: ground-penetrating radar; lunar exploration; Chang'e mission; *in situ* radar observation

1. Introduction

The Moon, Earth's nearest celestial body, has long captivated humanity, inspiring both myths and scientific inquiry. With advancements in modern technology, it has emerged as a strategic outpost for space exploration and a primary target for extraterrestrial studies (Benaroya 2017; Ding *et al.* 2022b). The Soviet Union ushered in a new era of lunar exploration in 1959 with the launch of Luna 1, the first successful lunar probe. A decade later, the Apollo 11 mission marked a historic milestone by achieving the first crewed lunar landing. After the Apollo program, lunar exploration stagnated for a period, however, the 21st century has witnessed a resurgence of global interest in returning to the Moon (Zheng *et al.* 2008; Jin *et al.* 2013). As lunar exploration progresses, ground-penetrating radar (GPR) has emerged as an essential tool in uncovering subsurface geological structures.

GPR is a non-invasive geophysical technique that generates continuous cross-sectional images of subsurface structures by transmitting microwave-frequency electromagnetic pulses and analyzing their reflections from materials with varying dielectric properties (Everett 2013). Recognized for its high resolution and broad applicability, GPR is particularly effective in planetary geology, especially for mapping shallow stratigraphy and ice deposits (Navarro & Eisen 2009). Beyond lunar applications, GPR has been successfully deployed on Mars and other celestial bodies to identify and investigate subsurface, ice thickness, and geological anomalies, providing invaluable data for planetary research (Boisson *et al.* 2011; Zhou *et al.* 2016; Zhang *et al.* 2022a).

This paper systematically reviews the development and significant achievements of lunar exploration missions equipped with GPR, highlights GPR's value in lunar scientific research, and synthesizes recent advances in radar data-processing techniques. We then assess the critical role that exploration radar will play in planning future lunar missions and outline anticipated development trends in GPR technology for lunar exploration.

Table 1 provides a concise summary of the instrument models and key technical specifications of the GPR systems deployed on a selection of representative lunar exploration missions. Detailed descriptions of each mission will be provided in Section 2.

2. GPRs in lunar exploration missions

2.1. Surface Electrical Properties experiment of Apollo 17

During the 1960s to 1970s, NASA conducted the Apollo program, a landmark initiative in lunar exploration that laid the foundation for advancements in lunar geological investigations and surface exploration technologies. Although GPR was not directly utilized in these early missions, various radar systems were extensively employed to investigate the lunar surface and shallow subsurface structures.

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Table 1. Technical specifications of GPR systems on lunar exploration missions.

Mission	Instrument	Central frequency	Penetration depth/vertical resolution	Sensing modality
Apollo 17	SEP Experiment	1, 2.1, 4, 8.1, 16, 32.1 MHz	Frequency-dependent	Surface *in situ* measurement
SELENE (Kaguya)	LRS	5 MHz	5000 m/~75 m	Orbital sounding
Chang'e 3–4	Dual-frequency LPR	500 MHz/60 MHz	500 MHz ~30 m/~0.3 m; 60 MHz ~100 m/~1 m	Surface *in situ* measurement
Chang'e 5–6	LRPR	2 GHz	4.5 m/~0.05 m	Surface *in situ* measurement

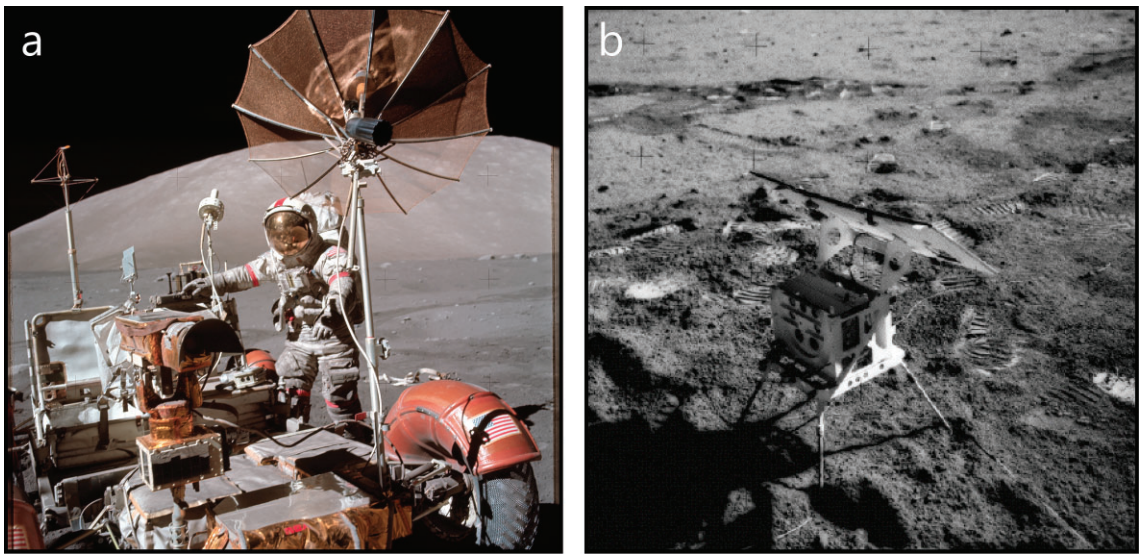

Figure 1. (a) The crewed lunar rover used during the Apollo 17 mission (https://www.nasa.gov/image-article/this-week-nasa-history-apollo-17-splashes-down-dec-19-1972/). (b) The instruments used in the SEP Experiment (https://catalog.archives.gov/id/16722835).

For instance, the Apollo 17 mission deployed a lunar surface radar system incorporating radio frequency interferometry, which is considered a precursor to modern GPR technology (Grimm Robert 2018).

The Apollo 17 Surface Electrical Properties (SEP) experiment, conducted during the Apollo 17 mission, utilized low-frequency electromagnetic waves (1–32 MHz) to measure the electrical conductivity and permittivity of the lunar regolith and shallow subsurface materials. As shown in Fig. 1b, the experiment employed two orthogonal horizontal electric dipole antennas on the lunar surface to transmit signals, while a triaxial electromagnetic receiver system mounted on the lunar rover (Fig. 1a) detected the emitted waves. By recording data on magnetic tapes and analyzing the interference patterns of electromagnetic wave propagation, the experiment revealed key electromagnetic characteristics of the lunar regolith (Simmons *et al.* 1973). The data obtained from this experiment significantly contributed to understanding the physical properties of the lunar surface and provided a critical foundation for future lunar exploration missions.

Despite its limitations in resolution and penetration depth, these pioneering radar systems established fundamental methodologies that later influenced the development of GPR technology and more advanced lunar geophysical instruments.

2.2. Lunar radar sounder of the Selenological and Engineering Explorer orbiter

In 2007, the Japan Aerospace Exploration Agency launched the Selenological and Engineering Explorer (SELENE) orbiter, later renamed Kaguya, with the primary objective of comprehensively investigating lunar geology and topography through high-resolution imaging and radar sounding (Kato *et al.* 2008). A principal instrument in this mission was the Lunar Radar Sounder (LRS), a specialized high-frequency radar system designed for subsurface structural analysis. Figure 2 shows the Kaguya orbiter in lunar orbit and highlights the location of the LRS antenna.

Operating within a frequency range of 10–30 MHz, the LRS was capable of probing subsurface structures at depths ranging from tens to hundreds of meters (Ono *et al.* 2008), making it particularly effective for studying lunar mare basalts and detecting deep geological formations (Ono *et al.* 2009). By analyzing radar reflections, scientists revealed key geological features such as stratified formations, lithological variations, and fracture networks.

Notably, the LRS detected subsurface lava tube structures (Kaku *et al.* 2017), which were later identified as promising candidates for future lunar habitats due to their natural shielding against cosmic radiation and micrometeoroid impacts. These discoveries have significant implications for the

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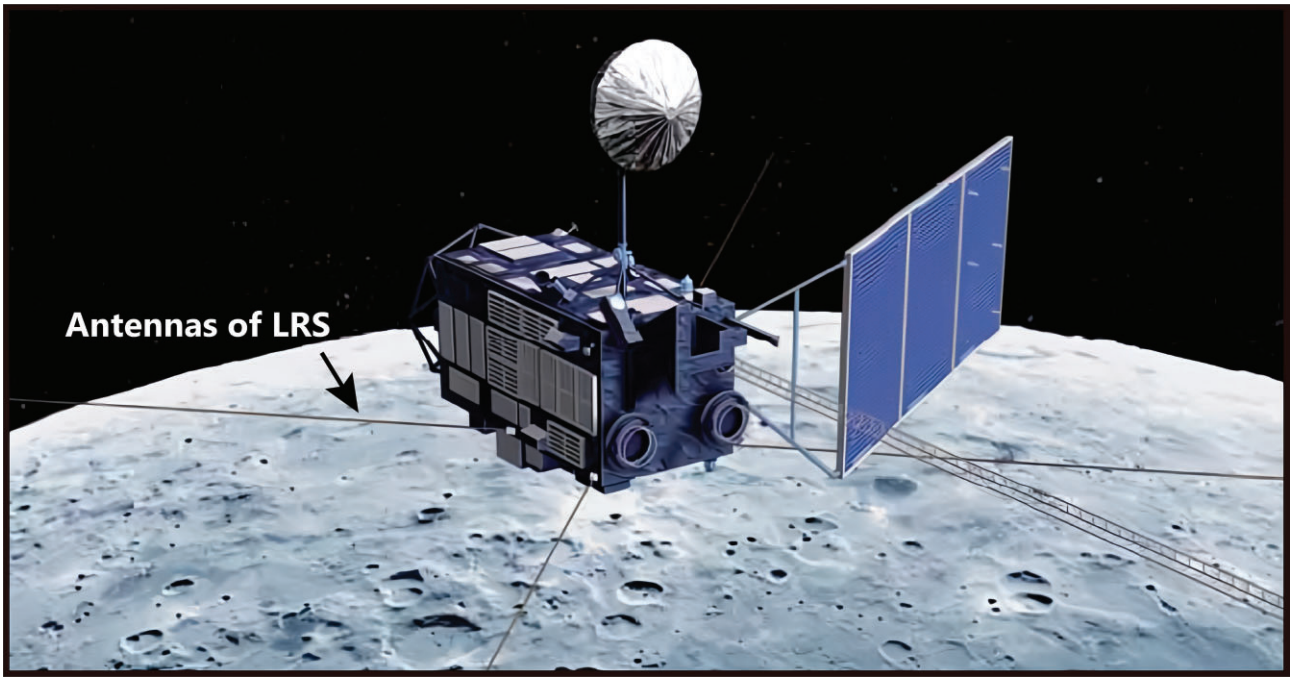

Figure 2. The surveyor used in the SELENE mission, with the LRS antennas indicated in the figure (https://www.kaguya.jaxa.jp/index_e.htm).

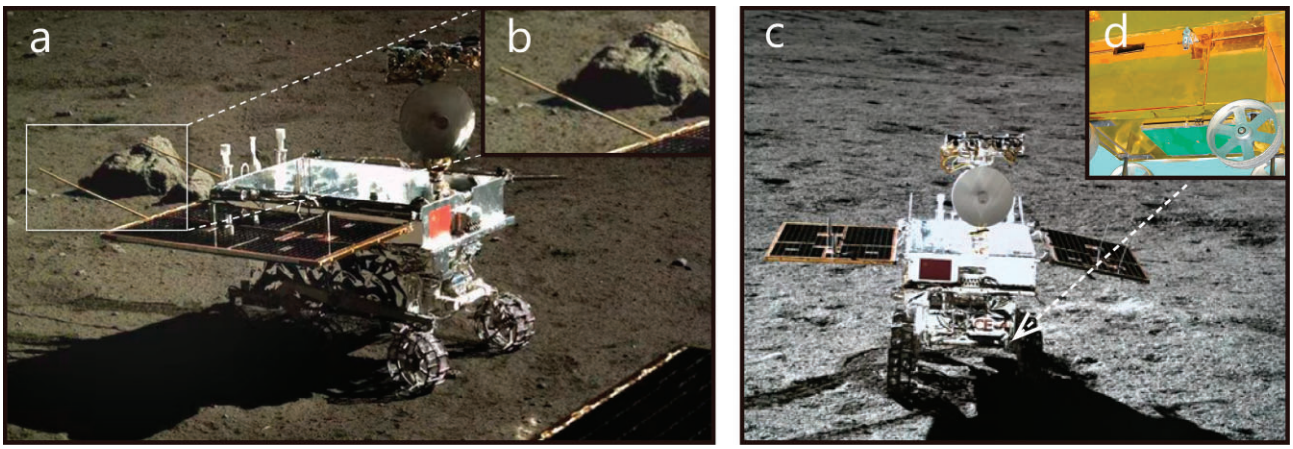

Figure 3. (a) The Yutu rover used in the Chang'e-3 missions (Image captured by the CE-3 lander, for details, please refer to the official website of the China National Space Administration), (b) low-frequency antennas, (c) the Yutu-2 rover used in the Chang'e-4 missions (Image captured by the CE-4 lander, for details, please refer to the official website of the China National Space Administration), and (d) high-frequency radar (Ding *et al.* 2021b).

selection of potential sites for future lunar bases, offering critical insights into the Moon's subsurface environment.

2.3. Lunar penetrating radar of Chang'e-3 and Chang'e-4

The Chinese Lunar Exploration Program (CLEP), also known as the Chang'e Project, is China's ongoing initiative in lunar exploration.

In 2013, the Chang'e-3 mission achieved China's first successful soft landing on the Moon, marking a major technological milestone. Referring to Fig. 3, the mission deployed the Yutu rover. Equipped with the lunar penetrating radar (LPR), a dual-frequency system, the mission aimed to probe both shallow and deep geological structures. The high-frequency radar (500–800 MHz), which is shown in Fig. 3d, primarily investigated the lunar regolith, reaching depths of several tens of meters and providing essential data on the thickness of the lunar regolith and the distribution of underlying rock layers at the landing site (Lai *et al.* 2016).

The low-frequency radar (40–80 MHz), which is depicted in Fig. 3b, penetrated hundreds of meters below the surface, revealing the thickness and distribution of subsurface lava layers. Radar results from Chang'e-3 identified distinct stratified structures beneath the lunar surface, where shallow material accumulation had been shaped by prolonged impact processes. The data also uncovered potential subsurface lava layers, reinforcing the hypothesis of extensive ancient volcanic activity and contributing to a deeper understanding of the volcanic evolution of the Mare Imbrium region (Zhang *et al.* 2015).

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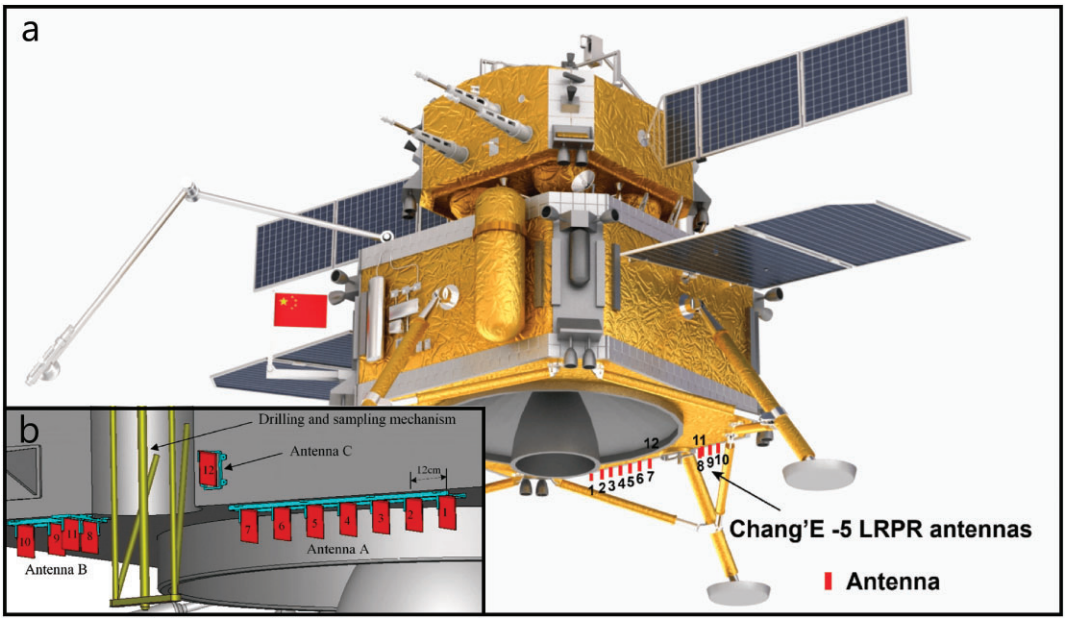

Figure 4. (a) The layout of the LRPR used in the Chang'e-5 and Chang'e-6 missions and (b) the positions of the positions of the LRPR's antennas, with a typical spacing of 12 cm between them (modified from Fang *et al.* 2023).

In 2019, the Chang'e-4 mission accomplished the world's first soft landing on the lunar far side. Before this mission, all previous lunar soft landings—totaling 20—had been conducted on the near side, even though the far side had been observed remotely more than 30 times (Wu *et al.* 2019). Chang'e-4's successful landing expanded the scope of lunar exploration, enabling direct geological investigations of the Moon's far side for the first time.

Its high-frequency radar detected stratified deposits in the shallow subsurface, revealing a complex distribution of impact crater ejecta and shallow lava flows. Thickness constraints derived from radar data provided chronological insights into impact events (Zhang *et al.* 2021b). Meanwhile, the low-frequency radar identified stacked multilayer lava flows in the deep subsurface, offering crucial insights into the formation and evolutionary history of the lunar farside crust (Lai *et al.* 2020).

2.4. Lunar regolith penetrating radar of Chang'e-5 and Chang'e-6

In 2020, China successfully executed the Chang'e-5 mission, marking its first lunar sample return endeavor. Equipped with the Lunar Regolith Penetrating Radar (LRPR), the mission aimed to analyze subsurface geological structures at the landing site, ensuring that the sampling site was scientifically representative and geologically diverse. Figure 4a shows the Chang'e-5's LRPR, with Fig. 4b detailing the 12 cm spacing between the antennas. The LRPR operated at a central frequency of 2000 MHz with a bandwidth of 1–3 GHz, achieving a detection depth exceeding 2 m and a vertical resolution of 5 cm (Fang *et al.* 2023). Analysis of the regolith's internal structure identified high-density anomalies, likely associated with rock fragments or agglutinates, within the sampling zone (Feng *et al.* 2022), these findings have guided optimal drilling site selection while enhancing studies on regolith composition and spatial distribution.

Chang'e-5's LRPR results revealed a complexly stratified shallow regolith, characterized by multilayer deposits likely formed by sequential impact ejecta. These findings challenge the prevailing assumption that the shallow subsurface consists predominantly of fine-grained regolith. Subsurface geological features provided new evidence for past lunar volcanic activity and impact deposition history. Furthermore, signal analysis refined estimates of regolith thickness and conductivity distribution, offering critical insights into lunar geological evolution and resource allocation (Fang *et al.* 2023).

In 2024, the Chang'e-6 mission extended lunar exploration to the South Pole region, becoming the first mission to conduct a landing and sample return in this scientifically significant area. By retrieving deep crustal and potential mantle materials, CE-6 is expected to significantly enhance our understanding of the Moon's internal composition, thermal evolution, and the role of primordial accretionary materials following the giant impact event (Zeng *et al.* 2023). Additionally, given the Lunar South Pole's potential water ice deposits, LRPR was deployed to map subsurface structures, support drilling operations, and investigate the presence of volatile resources: an essential factor for future lunar exploration (Li *et al.* 2024).

2.5. GPR of future lunar exploration missions

Building on the progress of the Chang'e-6 mission, China is steadily advancing its lunar exploration objectives toward long-term sustainability. The Chang'e-7 mission, scheduled for launch in 2026, will target the Lunar South Pole—a region of strategic interest due to its potential water ice reserves and other essential resources for sustaining future lunar habitats. The Chang'e-7 rover will carry five advanced scientific payloads, including a panoramic camera, a rover magnetometer, a lunar Raman spectrometer, a LPR, and an *in situ* volatile measuring system. Additionally, a mini-flying probe equipped with a Lunar Soil Water Molecule Analyzer will be deployed to enhance volatile detection capabilities (Wang *et al.* 2024).

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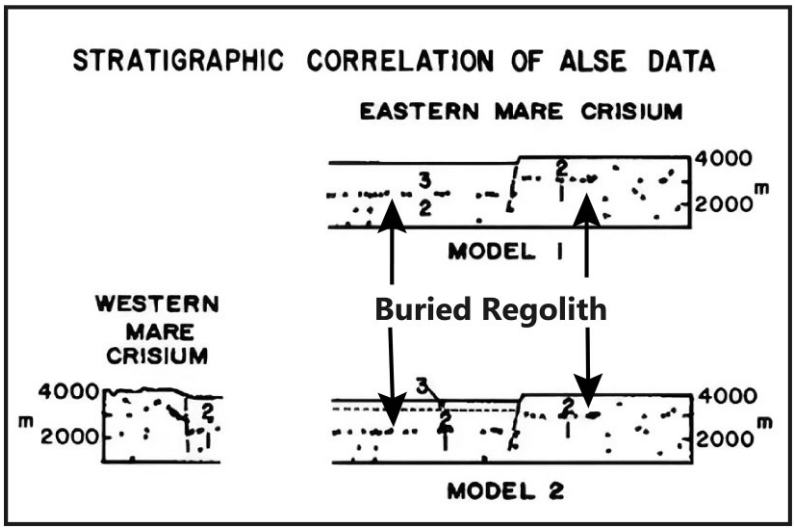

Figure 5. Stratigraphic models of the eastern and western Mare Crisium areas (modified from Maxwell & Phillips 1978). Total modeled thickness 4000 m, with Unit 1 reaching up to 2000 m. Arrows indicate stratigraphic boundaries corresponding to the buried regolith layer.

Scheduled for launch in 2028, the Chang'e-8 mission is designed to validate *in situ* resource utilization and 3D printing technologies crucial for lunar infrastructure development. The mission's LPR will support the identification of potential lunar construction materials and aid in optimizing landing-site selection, while also continuing to investigate subsurface water ice distributions—an essential resource for enabling sustained human presence on the Moon. Together, these missions will advance both lunar scientific knowledge and technological capabilities, laying the groundwork for future lunar habitation and exploration.

3. Scientific applications of GPR in lunar subsurface exploration

3.1. Identification of lunar subsurface stratigraphy

The first application of electromagnetic signals for subsurface stratigraphic identification on the Moon was conducted during the Apollo 17 mission (1972), utilizing the SEP and Apollo Lunar Sounder Experiment (ALSE) to investigate geological structures and generate subsurface profiles (Porcello *et al*. 1974).

ALSE employed a 5 MHz antenna expressly designed for the detection and location of subsurface discontinuities. By processing ALSE holographic data, Peeples *et al*. (1978) concluded that ALSE detected two subsurface radar reflectors with mean apparent depths of 0.9 and 1.6 km beneath the surface of Mare Serenitatis, as well as a single reflector at a mean depth of 1.4 km beneath Mare Crisium; all these are interpreted as basin-wide subsurface interfaces. Building on these observations, Maxwell & Phillips (1978) developed preliminary stratigraphic models for both the western and eastern margins of Mare Crisium. As illustrated in Fig. 5, in the western region of Mare Crisium, the subsurface reflector detected by the ALSE aligns with a surface slope break, consistent with normal faulting induced by mascon loading. Here, surface and subsurface displacements are nearly equivalent, supporting a simple tectonic fault model. In contrast, the eastern shelf exhibits a ~400 m vertical offset between the surface scarp and the ALSE reflector, prompting two alternative stratigraphic interpretations. The first interprets the reflector as the contact between Unit 2 and Unit 3, but this necessitates an abrupt thickening of Unit 2 to an implausible ~1000 m in the eastern shelf region, inconsistent with its annular morphology. The second and preferred interpretation identifies the reflector as the buried regolith contact between Unit 1 and Unit 2. This model limits Unit 3 to ~400 m thickness and aligns with independent constraints.

However, the subsequent Kaguya mission, which used antennas with the same center frequency, did not detect echo signals at those depths in the region. The subsurface reflectors detected by Kaguya were found at depths ranging from a few meters to several hundred meters (Ono *et al*. 2009; Oshigami *et al*. 2012). In light of advances in modern detection technology and data processing, as well as the high consistency of Kaguya's data, there is a possibility that the kilometer-scale deep reflectors reported by the early Apollo ALSE Experiment were overestimated. The Kaguya mission conducted large-scale subsurface exploration using the LRS, which transmitted and received electromagnetic waves to investigate subsurface structures. Operating within a frequency range of 1–30 MHz, with a central frequency of 5 MHz, the LRS exhibited strong penetration capabilities, enabling the detection of subsurface stratigraphy and potential water ice deposits beneath the lunar regolith and debris layers. Its penetration depth ranged from several hundred meters to many kilometers (Ono *et al*. 2008). Mounted on a polar-orbiting spacecraft, the LRS systematically scanned the lunar surface with a longitudinal step size of 1°, completing global coverage in ~15 days. As shown in Table 2, LRS data identified subsurface stratigraphy across Mare Serenitatis, Mare Smythii, Oceanus Procellarum, Mare Imbrium, and Mare Crisium, facilitating depth estimations and the calculation of loss tangent and relative permittivity values (Ono *et al*. 2009).

The Chang'e-3 (CE-3) mission represented China's first successful soft landing on an extraterrestrial body (Huang & Xu 2024). The spacecraft touched down in Sinus Iridum (44.12° N, 340.49° E), a basaltic lava plain northwest of Mare Imbrium, the exact landing site, indicated by the red

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Table 2. Summary of lunar subsurface horizons detected by LRS (Ono *et al.* 2009), The tangent value of the loss angle (tanδ) at the lowest layer is not determined. Here, *d* and *D* refer to apparent depth and estimated actual depth, respectively; ε represents the relative dielectric constant; and *P* denotes the echo power.

	Parameter	Procellarum (44.5° N, 69.3° W)	Imbrium (36.0° N, 15.3° W)	Crisium (57.0° N, 15.5° E)	Serenitatis (22.0° N, 23.3° E)	Smythii (1.0° N, 87.4° E)
Data samples		400	460	330	530	410
Top layer	tanδ	0.001–0.019	0.001–0.016	0.002–0.005	0.001–0.019	0.001–0.019
Second layer	*d* (m)	320	550	290	350	500
	D (m)	160	250	145	175	250
	P (dB)	−12.2	−12.7	−12.7	−11.4	−12.5
	ε	5.86–8.08	5.77–9.04	6.45–6.76	6.09–8.09	5.82–10.68
	tanδ	0.001–0.010	0.001–0.008	0.002		
Third layer	*d* (m)	770	1000	660		
	D (m)	343–347	454–460	290–293		
	P (dB)	−16.0	−15.6	−13.0		
	ε	7.64–11.03	7.52–11.89	7.52–11.89		

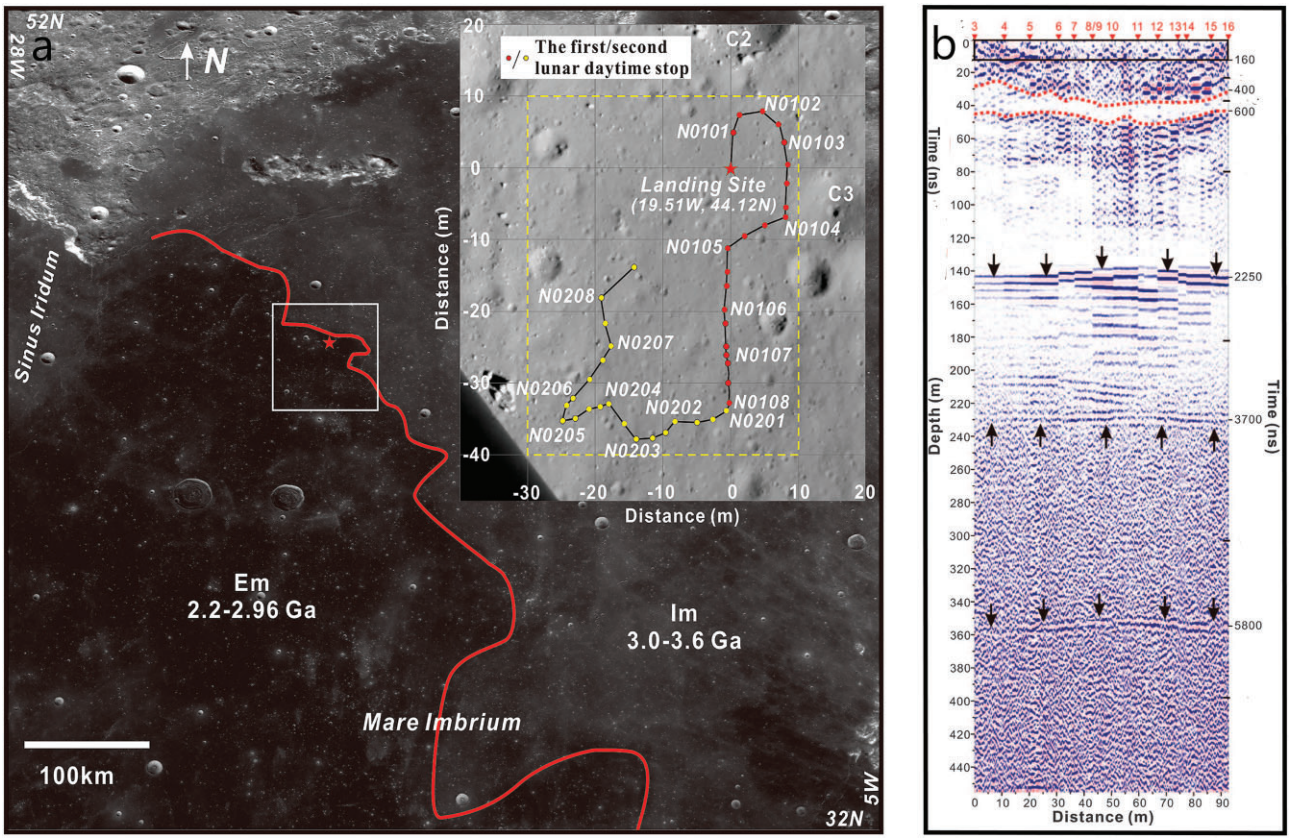

Figure 6. (a) The landing site of CE-3 mission on the lunar surface and the traversal route of Yutu lunar rover (modified from Yuan *et al.* 2020). (b) The low-frequency radar data (modified from Xiao *et al.* 2015).

star within the white box in Fig. 6a. The Yutu rover deployed during this mission carried the LPR, facilitating *in situ* subsurface structural investigations of the landing region.

Low-frequency LPR signals (60 MHz) enabled subsurface structural mapping down to ~400 m depth. As shown in Fig. 6b, the CE-3 landing site exhibits multiple stratigraphic units. The uppermost layers above 50 m depth are interpreted as ancient paleoregolith overlying Imbrian-period lava flows. Distinct radar reflectors were detected at depths of 140, 240, and 360 m. Notably, the layer between 140 and 240 m exhibits reflection characteristics indicative of stratified volcanic deposits, suggesting the presence of layered lava flows, pyroclastic deposits, or thin interbedded basaltic flows. These LPR observations indicate at least five discrete pyroclastic flow events, significantly refining our understanding of the complexity of lunar geological history (Xiao *et al.* 2015). Considering the presence of a 195 m-thick overlying basaltic unit, volcanic activity in the Imbrium Basin is

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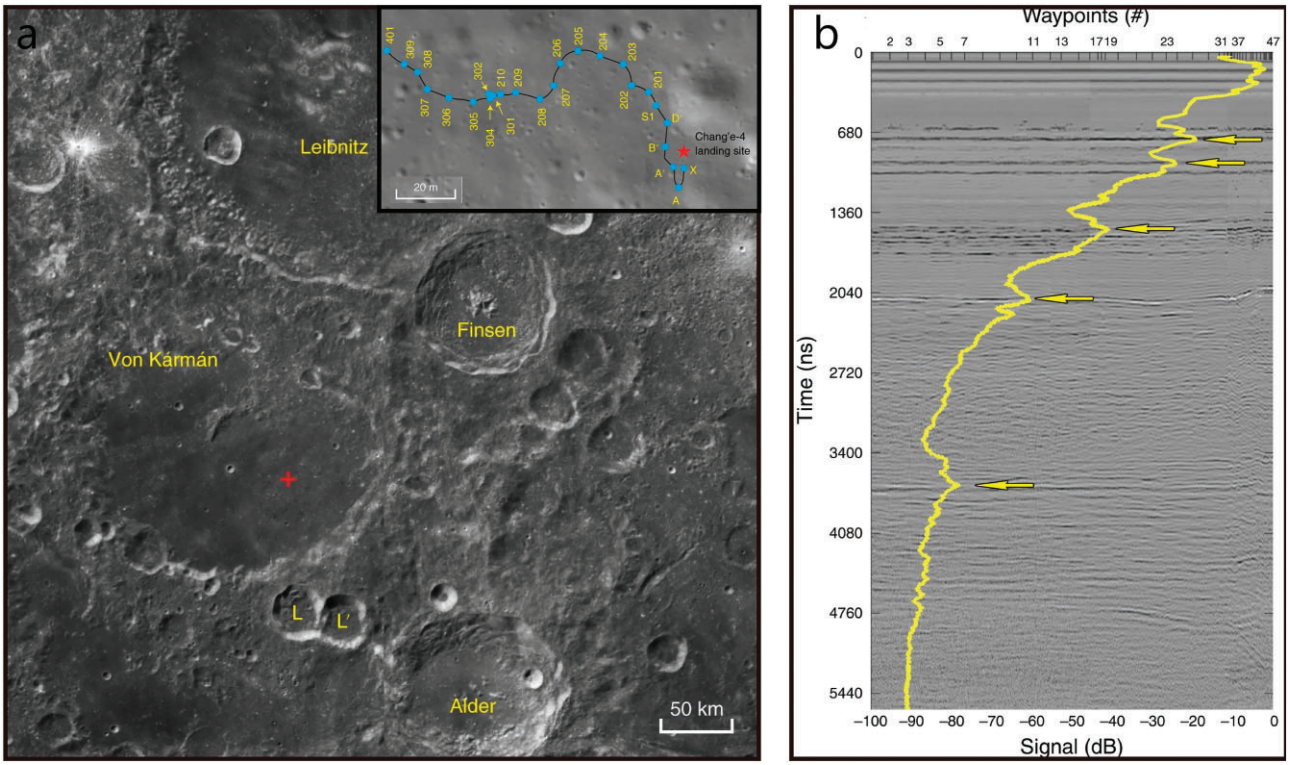

Figure 7. (a) The landing site of CE-4 mission on the lunar surface and the traversal route of Yutu-2 lunar rover (modified from Zhang *et al.* 2020a). (b) The low-frequency radar data (modified from Lai *et al.* 2020).

inferred to have persisted until ∼2.0–3.0 Ga (Zhang *et al.* 2015).

The Chang'e-4 (CE-4) mission landed on the lunar farside at 177.60° E, 45.44° S within Von Kármán crater (Liu *et al.* 2019), which is bordered by Leibniz Crater to the north, Finsen Crater to the east, and Alder Crater to the southeast. The Yutu-2 rover, equipped with an LPR identical to that of CE-3 (Lin *et al.* 2021), conducted *in situ* subsurface measurements along its traverse (Fig. 7). The extended traverse of Yutu-2 allowed for extensive data acquisition, providing valuable insights into the geological evolution of the lunar farside.

Low-frequency LPR (60 MHz) data identified three distinct subsurface geological units (Lai *et al.* 2020; Zhang *et al.* 2020a, 2020b; Yuan *et al.* 2021). In Unit I (∼0–130 m) are stratified mixtures of multistage lava flows and proximal crater ejecta. Layers between 50 and 110 m exhibit alternating horizontal radar reflections (Yuan *et al.* 2021), interpreted as Imbrian-period basaltic lava sequences (Zhang *et al.* 2020b). At a depth of ∼96 m, diverging stratigraphic orientations suggest that this unit was formed by ejecta from the Alder Crater to the east rather than by lava flows from the west (Lai *et al.* 2020). Numerical simulations of Alder Crater ejecta thickness support this interpretation, showing strong agreement with the observed stratigraphy at ∼110 m (Yuan *et al.* 2021). In Unit II (∼130–230 m), layered basalts are intercalated with potential paleoregolith horizons, indicating episodic volcanic activity. Unit III (>230 m) has thick (∼200 m) ejecta deposits attributed to the Leibniz Crater-forming impact, consistent with thickness predictions from crater scaling models (Zhang *et al.* 2020a). These findings provide key constraints on lunar volcanic and impact chronology, highlighting prolonged magmatic activity and intricate ejecta deposition processes.

Although substantial progress has been made in processing low-frequency LPR data from the CE-3 and CE-4 missions, the integrity of these observations remains contentious. Li *et al.* (2018) demonstrated that some of the deep radar features interpreted as shallow subsurface strata in the CE-3 LPR data are not true geological reflectors but rather signal artifacts produced by the radar system and its electromagnetic coupling with the metallic rover. This finding has since been corroborated by other researchers (Jia *et al.* 2019; Lei & Ding 2023). For example, Pettinelli *et al.* (2021) posited that in the low-frequency LPR recordings from the Chang'e-4 mission, only the 500–1800 ns window contains real, albeit faint, subsurface radar returns, and they argued that (Zhang *et al.* 2020a) had misidentified system noise as geological reflectors. In the reply (Zhang *et al.* 2021a), Zhang *et al.* acknowledged that the raw low-frequency radar data from CE-3 and CE-4 were dominated by electromagnetic interference and system noise. However, they demonstrated that by subtracting from each trace the mean response computed over several lunar days and by applying appropriately designed band-pass filters, the true subsurface reflection signals can be successfully recovered. Accordingly, low-frequency LPR data demand meticulous recalibration and validation.

3.2. High-resolution analysis of lunar regolith

Lunar regolith, also referred to as lunar soil, differs fundamentally from its terrestrial counterpart. Unlike Earth's regolith, which is shaped by atmospheric and biological processes, lunar regolith is primarily formed by continuous micrometeorite impacts, cosmic radiation exposure, and solar wind implantation (Langevin & Arnold 1977; Papike *et al.*

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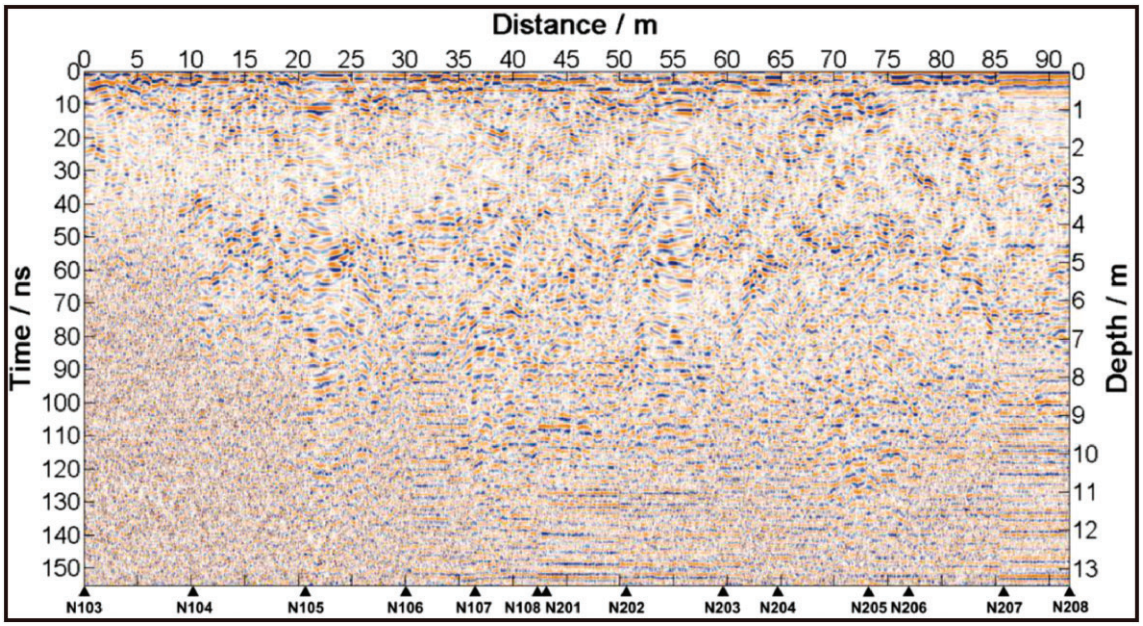

Figure 8. The processed results of high-frequency LPR data from the CE-3 mission (Zhang *et al.* 2018).

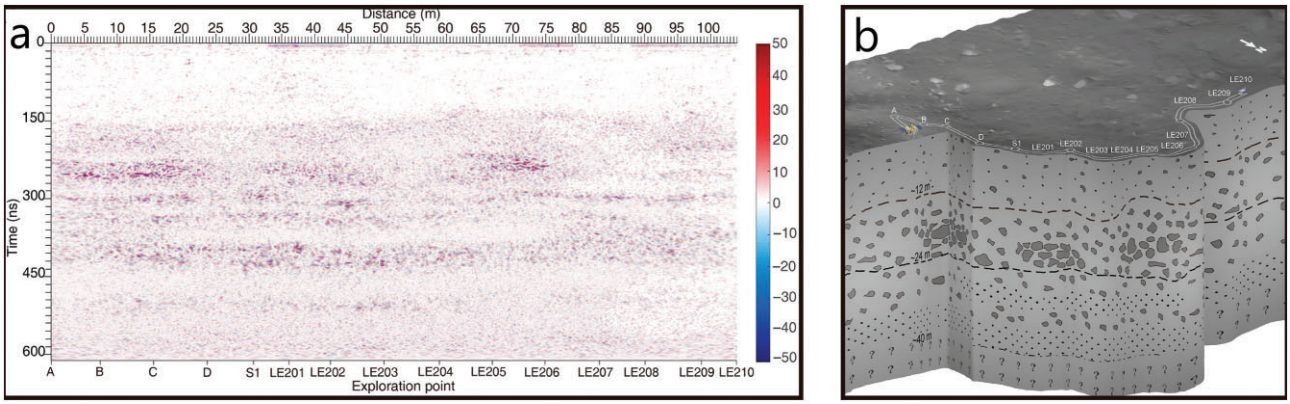

Figure 9. (a) The processed of high-frequency LPR data from the CE-4 mission. (b) The reconstructed underground stratigraphic structure, with the boundaries of the three main units marked at ∼12, 24, and 40 m depths (modified from CL Li *et al.* 2020).

1982; Taylor 1982). Analyses of Apollo-returned samples revealed a particle size distribution of 40 to 130 μm, with an average diameter of 70 μm (Carrier III *et al.* 1991). Previous investigations were constrained by the limited penetration depth of traditional methods. However, high-frequency (500 MHz) LPR onboard CE-3 and CE-4, along with LRPR (2000 MHz) on CE-5 and CE-6, has enabled new subsurface characterization.

Covering ∼114 m of traverse (Fig. 6), CE-3's Yutu rover conducted *in situ* analysis of the local regolith structure. High-resolution LPR data resolved four distinct subsurface layers (Fig. 8): Layer 1 (∼1 m depth) has fine-grained weathered regolith, shaped by micrometeorite fragmentation and cosmic irradiation. Its thickness exhibits spatial heterogeneity due to small-scale impact ejecta accumulation near the Ziwei Crater. In Layer 2 (1–4 m depth) are ejecta deposits from the Ziwei Crater, with thickness inversely correlated with distance from the crater rim. Layer 3 (4–10 m depth) has paleoregolith overlying ancient regolith materials (Xiao *et al.* 2015).

Subsequent studies corroborated this stratigraphic model, although discrepancies in layer interpretation persist. Some researchers subdivided Layer 1 into an upper, highly weathered sublayer and a lower, less altered sublayer, while others identified Layer 2 as a more altered paleoregolith (Zhang *et al.* 2019). Additional subsurface reflectors (2–5 events at <1-m depth) were attributed to localized impact ejecta (Fa *et al.* 2015). The regolith base exhibits an irregular morphology, with depths ranging from 2.2 to 5.4 m (Zhang *et al.* 2015). Dielectric permittivity-based classification delineates ejecta deposits (<3.25 m depth), including a weathered ejecta layer (<0.95 m), overlying a 4 m-thick lithic-soil admixture (Lai *et al.* 2016).

CE-4's landing site in the lunar farside Von Kármán crater (Fig. 7) contains geological units dating back ∼3.6 billion years. Yutu-2's high-frequency (500 MHz) LPR provided critical insights into regolith stratigraphy. As shown in Fig. 9b, the radar data reveals three principal units: The uppermost 12 m layer contains well-mixed ejecta materials sourced

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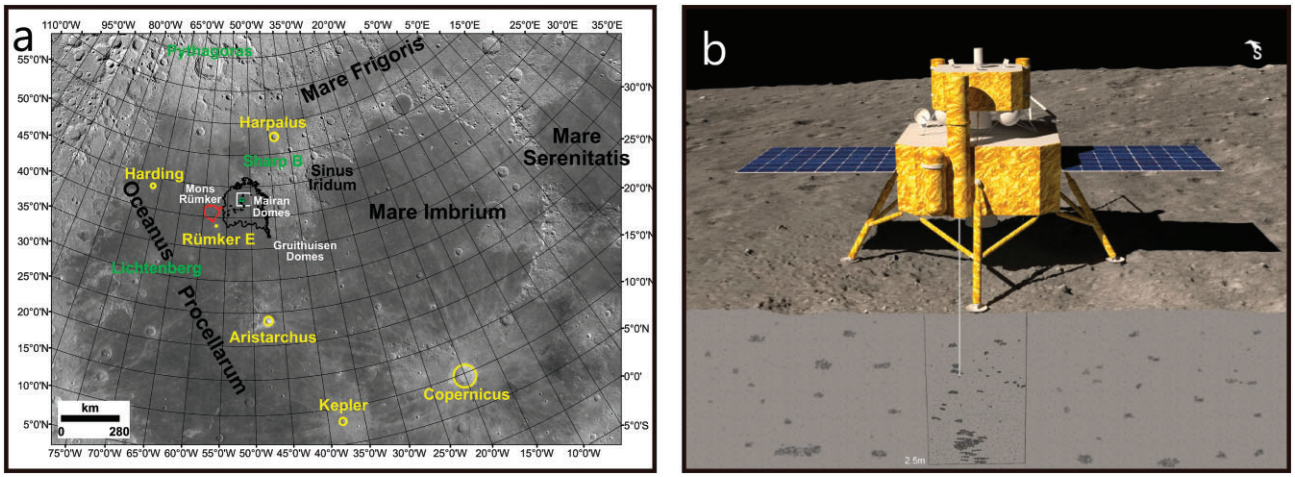

Figure 10. (a) The landing site of the CE-5 mission (modified from Qian *et al.* 2021). (b) The underground structural image obtained from LRPR detection, showing subsurface structures down to a depth of 2.5 m (modified from Su *et al.* 2022).

from Finsen, L, and L' impact events. The intermediate layer (12–24 m) displays boulder clusters and thin fine-grained interlayers, possibly resulting from impact-induced shearing or inter-event weathering processes. The third layer (24–40 m) exhibits rhythmic bedding with components predating Finsen crater, including a distinctive fine-grained lenticular structure at ∼28 m depth (Li *et al.* 2020b).

An alternative hypothesis posits that the Finsen ejecta deposit extends to a depth of 34 m (Zhang *et al.* 2020a). Meanwhile, Lai *et al.* (2019) proposed that Unit III represents a homogeneous, compact basaltic substrate. With the continuous accumulation of high-frequency radar data from the CE-4 mission, new breakthroughs have been achieved in studies of the shallow subsurface structure at the landing site. Early analyses of high-frequency radar data from the first two lunar days of the mission provided preliminary constraints on the dielectric properties of the high relative permittivity buried lens at ∼27 meters beneath the lunar surface. Combined with regional geological context and volcanic history, this layer was interpreted as a buried Eratosthenian basaltic lava flow (Ding *et al.* 2023). Further analysis by Giannakis *et al.* (2024a), utilizing high-frequency radar data collected along the first 1000-meter traverse of the Yutu-2 rover, identified multilayer Imbrian basaltic lava flows through frequency dispersion modeling and reflection signatures within the 10–30 m depth range. These findings not only refine the volcanic stratigraphic model of the Von Kármán crater but also offer critical insights into the spatiotemporal distribution of Imbrian volcanic events on the lunar farside. Building on prior studies, Zhang *et al.* (2024a) employed high-frequency radar data from the Yutu-2 rover's 1500-meter traverse to identify an 8-meter-thick paleoregolith layer at ∼10 meters below the lunar surface, predating the Finsen impact event. Radar measurements revealed that this layer exhibits lower relative permittivity and absence of high-Titanium signatures—sharp contrasts to the underlying Imbrian basalt flows. Subsequently, Soldovieri *et al.* (2025) applied a microwave tomographic imaging approach to the Chang'e-4 mission's 500 MHz high-frequency LPR data—collected over 53 lunar days along a 1340 m traverse by the Yutu-2 rover—to generate high-resolution subsurface images down to ∼40 m in depth.

CE-4's regolith also contains deep-seated materials excavated by large impacts, potentially originating from the upper lunar mantle (Gou *et al.* 2019). LPR observations facilitate a deeper understanding of lunar geological processes and impact histories, aiding in the identification of deep-sourced material enrichment zones. However, further advancements in inversion modeling are required to precisely determine the provenance and composition of these geological units.

The LRPR system onboard CE-5 operates at a central frequency of 2000 MHz, providing high-resolution imaging for precise stratigraphic characterization and optimized core sampling strategies. The CE-5 landing site is located on an Eratosthenian-aged mare basalt plain characterized by a gentle topography, with an average slope of 2.7 m over a 120 m baseline (Wu *et al.* 2018). More than 90% of the surrounding terrain exhibits inclinations of $<5^{\circ}$ (Qiao *et al.* 2021).

Estimates from the Lunar Reconnaissance Orbiter Camera on board the Narrow Angle Camera suggest that the regolith thickness at the CE-5 landing site ranges from 4 to 7 m (Qian *et al.* 2021), This estimate is corroborated by subsequent measurements indicating a range of 1.5 to 8 m, with a median thickness of ∼5 m (Jia *et al.* 2021). LRPR datasets facilitated stratigraphic delineation and source tracing of regolith materials down to a depth of 2.5 m at the CE-5 drill site. Radar stratigraphy identifies the upper 80 cm as a heterogeneous mixture of Xu Guangqi Crater ejecta and locally derived mare basalt weathering products. The 80–100-cm interval consists of pre-Xu Guangqi Crater mare basalt deposits of Eratosthenian age (Fig. 10b), representing autochthonous bedrock materials (Su *et al.* 2022).

3.3. Identification of special subsurface structure on the Moon

The Moon's subsurface harbors a diverse and complex geological structure, which not only preserves records of early volcanic activity and intense meteoritic impacts but also provides crucial insights into the Moon's internal evolution and tectonic reshaping. Various lunar exploration missions have detected unique subsurface structures, including lava tubes, ancient impact craters, and bifurcated structure.

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Lava tubes are hollow tunnels formed when the surface of a flowing lava stream cools and solidifies, while the molten lava inside continues to flow until it is completely drained. These structures are considered prime candidates for future lunar base construction due to their potential for providing natural shielding against radiation and micrometeorite impacts (Duke *et al*. 1985). In 2009, data from the Terrain Camera and Multiband Imager onboard the Kaguya spacecraft revealed a skylight ∼65 meters in diameter on the surface of Marius Hills. This feature is believed to be a collapse pit formed by the partial collapse of an underlying lava tube (Haruyama *et al*. 2009). Further evidence supporting this finding came from Kaguya's LRS data, in conjunction with gravity anomaly measurements (Kaku *et al*. 2017). However, the LRS operates at a frequency of 2 MHz, which corresponds to a resolution of 75 meters in free space. This resolution limits its ability to detect lava tubes that may exist tens of meters beneath the lunar surface. Subsequent studies used Kaguya LRS data to establish criteria for identifying lunar lava tube floors. Ambiguity tests indicated that the signals previously attributed to lava tubes were, in fact, cluttered surface echoes rather than direct reflections from subsurface lava tubes (Kobayashi *et al*. 2021).

The LPR used in the CLEP offers a significant resolution advantage over the LRS, allowing for more precise *in situ* detection of potential lava tubes along the survey line. When an electromagnetic wave propagates from a high-dielectric material to a low-dielectric material and then back to a high-dielectric material, its reflected wave undergoes a phase reversal (Neal 2004). Consequently, when radar waves enter a lava tube through its roof, travel through the cavity, and then reach the tube's floor, the reflected wave exhibits a phase inversion. Based on this principle, GPR successfully identified the Koumoriana lava tube beneath Mount Fuji, accurately reconstructing its size and depth (Miyamoto *et al*. 2005). Applying the same methodology, high-frequency radar data from the Chang'e-3 mission revealed subsurface cavities, interpreted as accumulations of basaltic boulders resulting from ejected deposits (Ding *et al*. 2021a). Although the detection of lunar lava tubes and other subsurface cavities remains in its early stages, GPR has proven successful in identifying lava tubes on Earth, yielding numerous valuable findings (Rowell *et al*. 2010; Esmaeili *et al*. 2020). With continuous advancements in radar technology and improvements in resolution, particularly the development of multiband penetrating radar techniques, scientists will gain deeper insights into the Moon's geological processes (Carrer *et al*. 2018). This progress will also provide more precise guidance for selecting future lunar base locations.

Impact craters are the most prominent landforms on the Moon. The Narrow Angle Camera aboard the Lunar Reconnaissance Orbiter Camera has captured extensive high-resolution images of lunar craters, providing crucial data for studying the Moon's regolith. However, gravity data from the GRAIL mission indicate that numerous ancient impact craters remain buried beneath the lunar regolith (Evans *et al*. 2016). Due to the resolution limitations of gravimetry, only large ancient craters have been detected.

With the successful landing of the Chang'e-4 mission on the far side of the Moon, high-resolution GPR data have enabled the identification of smaller subsurface impact craters. CE-4's LPR data also revealed a significant impact signature at ∼10 m depth, interpreted as a meteoroid that penetrated through four pre-existing ejecta layers (Zhang *et al*. 2021b). This interpretation is supported by subsequent investigations (Feng *et al*. 2022). Furthermore, GPR analysis has revealed that deep-seated rocks exhibit plastic deformation due to impact forces, shedding light on the structural limits of rock failure. Additionally, the radar data uncovered an ancient buried lunar surface (Lai *et al*. 2021). Radar-detected concave features within the upper 35 m of regolith are interpreted as preserved impact structures, enabling reconstruction of stratigraphic deposition rates and geological evolution (Yuan *et al*. 2021). Recent studies have also identified a flat-bottomed crater beneath Von Kármán crater at a depth of ∼15 meters, which is believed to have formed due to a secondary low-velocity impact within a pre-existing crater (Zhang *et al*. 2024b). Additionally, low-frequency GPR data from CE-4 have revealed a branching subsurface structure beneath Von Kármán crater, which is hypothesized to have originated from ejecta deposits from the Schrödinger basin (Cao *et al*. 2023).

As GPR technology and other lunar exploration techniques continue to advance, our ability to investigate these unique subsurface structures will significantly improve. This progress will offer deeper insights into the Moon's geological evolution and developmental history.

4. Advancements in lunar penetrating radar data processing

The vast amount of radar data acquired from various lunar missions has driven significant advancements in LPR data processing methodologies. The Moon's distinct geological conditions pose unique challenges compared to terrestrial applications, necessitating the development of specialized processing techniques. For instance, leveraging CE-4's 500 MHz LPR data, Roncoroni *et al*. (2024a) proposed a streamlined pre-processing framework for Chang'e-4, encompassing trace-cleanup, coordinate reprojection, format conversion, and a unified five-step signal-processing chain. In addition, lunar radar data processing has evolved across multiple dimensions, with innovations specifically tailored to the Moon's complex subsurface environment.

4.1. Forward modeling of lunar regolith

Forward modeling involves simulating radar signal propagation and reflection processes in subsurface media using electromagnetic wave theory, based on predefined geological structures and physical parameters. In lunar radar data processing, this approach is primarily employed to validate data interpretation, optimize radar system designs, and support inversion analysis. By developing electromagnetic models of lunar regolith, researchers can simulate radar signal propagation and reflection behaviors, offering crucial references for interpreting observational datasets. These simulation efforts provide essential guidance for radar data processing and analysis.

To address the scattering of radar-transmitted electromagnetic waves in lunar regolith, a scattering model based on vector radiative transfer theory in random media was developed. This model represents lunar regolith as a homogeneous fine-grained medium with rough parallel interfaces and embedded dielectric inclusions, characterizing five distinct scattering mechanisms (Fa *et al*. 2011). A multilayer parallel regolith model, excluding rock components, was established

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to support the design of the CE-3 radar system. This model facilitated the analysis of wave propagation, attenuation, reflection, geometric spreading, and antenna directivity (Fa 2013).

Following CE-3's successful landing, refined models were developed to enhance LPR data interpretation. A multilayer regolith simulation framework was introduced to better replicate actual lunar subsurface structures. This model stratifies the lunar surface into four distinct layers: vacuum, regolith, ejecta (impact deposits), and bedrock. By comparing simulated radar B-scan images with CE-3 observational data, researchers quantitatively determined regolith thickness through systematic parameter optimization. Additionally, a dielectric permittivity model based on the clipped Gaussian random field theory was developed, enabling precise reconstruction of regolith material distribution via inversion techniques (Li *et al*. 2017). A more sophisticated 3D regolith model was subsequently constructed, incorporating critical factors such as terrain variations, rock inclusions, medium stochasticity, and depth-dependent permittivity gradients. This model successfully interpreted CE-3 LPR datasets (N201-N203 profiles), allowing for quantitative estimations of regolith thickness, localized rock distributions, and subsurface heterogeneity (Zhang *et al*. 2018).

The CE-4 mission provided additional data for lunar regolith studies, facilitating the integration of CE-3 and CE-4 datasets for comparative analysis. Lv *et al*. (2020) conducted forward modeling to analyze potential subsurface reflectors using this combined dataset. To improve regolith simulation fidelity, Wang *et al*. (2021) introduced a model incorporating horizontal stochastic perturbations and depth-dependent permittivity gradients. Further refinements included the addition of discrete stratified structures, with comparative analysis against CE-3 data revealing stratified features within the upper ~4 m of regolith (Ding *et al*. 2024).

Forward modeling plays a pivotal role in LPR data processing. By constructing multilayer electromagnetic models of lunar regolith, scientists can effectively predict radar signal propagation and reflection characteristics, validate radar system optimizations, and assist inverse data analysis. Despite continuous advancements in model complexity and precision, challenges persist in addressing medium parameter uncertainties and inherent model simplifications. Future advancements integrating expanded lunar exploration datasets with refined simulation methodologies, are expected to improve model accuracy and predictive capabilities, thereby furthering our understanding of the lunar geological framework and advancing LPR technological development.

4.2. Regarding the determination of dielectric property parameters for lunar regolith

The LPR records the two-way travel time of electromagnetic signals rather than directly measuring the physical depth of subsurface features. Accurate stratigraphic depth determination within the lunar regolith from radar signals requires prior knowledge of its dielectric properties, with the dielectric constant being the most crucial parameter. The dielectric constant is a complex quantity consisting of a real component (relative permittivity), which governs the velocity of electromagnetic waves in the medium, and an imaginary component, which influences signal attenuation. The ratio of the imaginary to real components defines the loss tangent.

Initial estimations of the relative permittivity of lunar regolith date back to the Apollo era. Laboratory analysis of Apollo-returned regolith samples indicated a relative permittivity range of 3–6 (Carrier III *et al*. 1991). These results, derived from localized sampling sites, may reflect site-specific characteristics and could be influenced by potential alterations introduced during laboratory analysis. The Kaguya mission later provided a more extensive dielectric characterization using LRS data, offering permittivity estimates across different lunar regions (Ono *et al*. 2009) as summarized in Table 2.

The Chang'e lunar exploration program enabled higher-precision dielectric characterization through *in situ* LPR measurements near landing sites. Early LPR data processing utilized dielectric constants estimated from FeO and TiO_2 abundance measured by the Lunar Prospector gamma-ray spectrometer, combined with regolith porosity models and empirical permittivity-composition relationships, yielding an estimated value of $3 + i0.03$ (Fa *et al*. 2015). A widely adopted approach for estimating dielectric constants from GPR data involves hyperbolic curve fitting. When LPR-transmitted radar waves encounter subsurface discontinuities, such as boulders or voids, partial reflections occur while the remaining waves propagate further. If the antenna moves perpendicularly across the object's axis, hyperbolic reflection patterns emerge in the radargram (Ristic *et al*. 2009). By analyzing these hyperbolas, the average electromagnetic wave velocity at a given depth can be determined, thereby allowing for the calculation of relative permittivity. High-frequency LPR data from CE-3 indicated an average relative permittivity of 2.47 at a depth of ~3 m, aligning with Apollo-era estimates (Lai *et al*. 2016), while values increased to ~8 at depths >3 m (Feng *et al*. 2017). CE-4 data also employed hyperbolic fitting, with 823 hyperbolas extracted from high-frequency LPR data, yielding regolith permittivity values between 2.2 and 5.0 (Dong *et al*. 2020a).

Laboratory studies have established a correlation between relative permittivity and bulk density in geological media. High-frequency LPR measurements from Chang'e-4 were used to estimate regolith density, yielding values of 1.90 ± 0.08 g/cm³ or 1.67 ± 0.07 g/cm³ at depths exceeding 5.8 m via permittivity-density conversion models (Li & Zhang 2021). A novel low-data-dependency processing framework was introduced to estimate dielectric properties and subsurface anomalies by leveraging electromagnetic wavefront propagation paths, eliminating reliance on prior assumptions (Zhang *et al*. 2018). Additionally, a refined approach employed normalized velocity spectra derived from cross-channel similarity analysis of dual high-frequency CE-4 datasets, enabling spatial permittivity mapping through velocity interpolation techniques (Song *et al*. 2021). Many of these methodologies required manual selection of hyperbolas in radar data, introducing potential subjective biases. To address this limitation, automated hyperbola detection was implemented using three-dimensional normalized velocity spectra. Its application to CE-4 LPR datasets enabled robust automated parameter extraction, revealing previously unobserved subsurface stratigraphic discontinuities (Dong *et al*. 2020b). Beyond hyperbolic fitting techniques, relative permittivity can also be inferred from radar signal amplitude. By analyzing LPR transmission amplitudes calibrated through ground-based experiments and comparing them with *in situ* reflection signal intensities, regolith permittivity

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was estimated via amplitude ratio analysis (Dong *et al.* 2020a).

The Chang'e-5 mission provided the most precise dielectric measurements to date by integrating returned regolith sample analysis with *in situ* LRPR observations. Laboratory measurements of CE-5 regolith samples yielded a relative permittivity of 3.04, consistent with Apollo-derived estimates. However, *in situ* LRPR-derived values were slightly lower at 2.48, with the discrepancy attributed to compaction effects in laboratory sample preparation (Su *et al.* 2022).

As lunar regolith constitutes a lossy propagation medium, electromagnetic waves experience energy attenuation during subsurface transmission. The loss tangent (tanδ) quantitatively characterizes electromagnetic energy dissipation in such media. Laboratory analyses of Apollo samples established regolith loss tangent values within the range of 5–9 $\times$ 10^{-3} (Carrier III *et al.* 1991). *In situ* SEP measurements from Apollo 17 suggested a broader range of 3–18 $\times$ 10^{-3} (Grimm Robert 2018). The Kaguya mission further provided regional tanδ estimates across various lunar terrains, as detailed in Table 2.

Ground validation experiments for Chang'e-3's LPR were conducted at Laohugou Glacier in the Qilian Mountains, Gansu Province, chosen for its desiccated basaltic terrain, which closely resembles lunar surface conditions. This glacial medium exhibits dielectric properties (ε_r = 3.2, tanδ = 5–10 $\times$ 10^{-3}) comparable to those of lunar regolith. Calibration tests demonstrated that the high-frequency LPR could resolve englacial structures down to ~50 m with a vertical resolution of ~17 cm (Zhang *et al.* 2014). However, operational data from actual lunar deployments revealed contrasting results: CE-3's high-frequency LPR achieved a penetration depth of only ~10 m, while CE-4's LPR extended to ~40 m. This discrepancy in radar wave attenuation between the lunar nearside and farside is primarily attributed to variations in TiO_2 + FeO content, which influence electromagnetic absorption characteristics. The precise determination of regolith tanδ is critical for optimizing LPR system performance. The radar equation-based inversion approach allows for precise quantification of tanδ by analyzing signal attenuation characteristics. By integrating Kaguya Multiband Imager Fe-Ti abundance maps (Qiao *et al.* 2019) with CE-4 radar data, researchers estimated the tanδ at the CE-4 landing site to be in the range of 0.004–0.005, while radar-based inversion constrained the farside regolith FeO + TiO_2 content to 11.8–13 wt% (Lai *et al.* 2019).

Applying the frequency-offset method to CE-3 high-frequency radar data processing, the results reveal that the loss tangent (tanδ) of the lunar regolith within the upper ~4 m ranges from ~11 to 17 $\times$ 10^{-3} (Ding *et al.* 2020). Based on the established correlation between loss tangent and penetration depth in high-frequency LPR (Xing *et al.* 2017), and assuming a relative permittivity of $\varepsilon_r = 2.9$, the corresponding high-frequency LPR penetration depth was estimated to be 12.5–15.7 m. The Fe-Ti abundance derived from radar-based tanδ estimations (24–30 wt%) was found to be consistent with *in situ* APXS measurements of 27.8 wt% conducted by the CE-3 rover. Furthermore, laboratory analysis of regolith samples returned by the CE-5 mission (FeO 22.7 wt%, TiO_2 5.09 wt%) corroborated the radar-derived compositional estimates (Yi *et al.* 2023). The greater penetration depth achieved by CE-4's LPR can be primarily attributed to: (i) lower Fe-Ti oxide content (11.8–13 wt%) at the CE-4 landing site, reducing electromagnetic attenuation, and (ii) lower near-surface boulder abundance and a thicker fine-grained regolith layer, which minimized scattering losses. These combined factors account for the significant disparity in subsurface penetration depth observed between the two landing sites.

4.3. Research on imaging methodologies for GPR data

The mathematical similarity between electromagnetic wave propagation and seismic wavefield dynamics enables the adaptation of advanced seismic imaging techniques to GPR data processing. Wave-equation migration algorithms—including Kirchhoff migration, reverse-time migration (RTM), and phase-shift migration—enable high-resolution subsurface imaging by repositioning reflection interfaces to their correct geological locations and refining depth estimations (Wang *et al.* 2023). These methods play a crucial role in processing LPR datasets by mitigating geometric distortions in reflection signals and enhancing the interpretability of subsurface features (Liu *et al.* 2023). Migration-enhanced processing of Chang'e-3 and Chang'e-4 radar datasets significantly improved the resolution of subsurface reflectors and their spatial distributions. By separating LPR signals into reflection and diffraction components, applying localized diffraction focusing, and performing migration imaging, researchers achieved more accurate velocity models and enhanced imaging fidelity (Li & Zhang 2021). A hybrid approach integrating frequency-wavenumber (FK) migration and minimum entropy optimization has been proposed to overcome the limitations of traditional hyperbolic fitting in GPR velocity estimation, particularly in cases where wave diffraction is minimal (Chen *et al.* 2022). Furthermore, a hierarchical imaging framework employing RTM combined with finite-difference time-domain simulations was introduced for CE-4 radar data, allowing for improved lithological boundary delineation (Giannakis *et al.* 2022). Numerical modeling incorporating factors such as near-surface fractures, heterogeneous media, and complex topography at the CE-4 landing zone demonstrated that iterative diffraction wave migration with varying velocities significantly improved fracture pattern identification (Huo *et al.* 2023). To mitigate diffraction contamination in conventional Kirchhoff migration (KM), a modified least-squares Kirchhoff migration algorithm was introduced, effectively suppressing noise while preserving lateral variations in stratigraphic continuity (Ding *et al.* 2023). However, some studies highlight that migration-induced edge effects introduce interpretational challenges in deep subsurface imaging (Forte *et al.* 2023).

Advanced migration imaging techniques have been specifically developed for the LRPR on Chang'e-5 and Chang'e-6, leveraging its higher operating frequencies to enhance subsurface imaging capabilities. Two-dimensional RTM, implemented via frequency-domain asymptotic data transformation filters, was successfully validated using controlled LRPR measurements in volcanic ash analog environments (Liu *et al.* 2017). A frequency-domain RTM algorithm incorporating dyadic Green's functions for layered media was later introduced, enabling rapid 3D reconstruction of subsurface lithic structures (Liu *et al.* 2018). Additionally, prestack depth migration techniques were applied for detailed stratigraphic imaging (Li *et al.* 2018b). To further refine imaging accuracy, envelope velocity spectral analysis was employed to improve electromagnetic wave velocity models, integrated

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with Hilbert-transform-based signal processing to enhance prestack migration resolution (Feng *et al*. 2019). Following successful implementation of 3D high-fidelity RTM on LRPR datasets (Zhang *et al*. 2019), the pseudospectral time-domain method was incorporated, significantly improving imaging accuracy in regions with high subsurface uncertainty (Zhang *et al*. 2022b). Given the non-uniform antenna geometry of LRPR and the presence of stratified subsurface lithology at the CE-5 landing site, a prestack depth migration method based on Kirchhoff integral operators was developed to optimize imaging performance (Ding *et al*. 2022a). Moreover, microwave tomography was applied to LRPR data processing, with validation from synthetic simulations and terrestrial lunar regolith analog datasets confirming its effectiveness (Catapano *et al*. 2023).

The rise of machine learning (ML) techniques has introduced transformative applications in GPR imaging and interpretation. Several studies have demonstrated ML's effectiveness in terrestrial GPR applications (Giannakis *et al*. 2019; Sun H. *et al*. 2022; Dai *et al*. 2023; Patsia *et al*. 2023). However, the lack of labeled data remains a major limitation for lunar applications. To address this, an unsupervised ML framework was developed for automated radar facies classification in CE-4 datasets, successfully identifying a previously unrecognized basaltic layer (Giannakis *et al*. 2024b). Furthermore, deep learning methodologies have been employed to reconstruct ancient regolith layer geometries beneath the CE-4 landing site, offering new insights into lunar subsurface stratigraphy (Roncoroni *et al*. 2024b).

5. Summary and prospects

GPR technology has played a pivotal role in advancing lunar exploration, significantly enhancing our understanding of the Moon's subsurface structure, regolith properties, and resource distribution. From the pioneering Apollo program to the latest Chang'e missions, GPR systems have provided invaluable datasets that have deepened our knowledge of lunar geological processes, surface evolution, and planetary differentiation. However, comparing these radar sounders and their processing workflows reveals key trade-offs. Apollo 17's ALSE employed a 5 MHz subsurface sounder antenna with limited bandwidth, yielding only coarse vertical resolution and increasing the risk that random noise spikes or surface clutter would be misidentified as deep, kilometer-scale reflectors. By contrast, Kaguya's LRS, operating over the same central frequency, used automatic gain control, multi-orbit coherent stacking and clutter suppression. These enhancements provide finer range resolution and more consistent echo detection, but in practice confine reliable subsurface horizons to the upper few hundred meters. Meanwhile, Chang'e-3's LPR achieves sub-meter vertical resolution, while Chang'e-5's LPRP offers centimeter-scale resolution. Although these systems excel at characterizing the upper tens of meters of regolith and volcanic deposits, they lack the penetration needed to validate kilometer-scale interfaces. Ultimately, a comprehensive view of the lunar subsurface requires fusing multi-frequency datasets and applying advanced processing techniques such as adaptive digital filtering, coherent stacking, and tomographic reconstruction, to capitalize on each system's strengths while mitigating their individual limitations.

Major spacefaring nations, including China, the USA, and European partners, have proposed establishing permanent lunar bases. In this context, lunar GPR will play an essential role in both landing-site selection and *in situ* resource evaluation. The Chang'e-7 and ESA Prospect missions, each targeting the Lunar South Pole, will deploy GPR to map the distribution of water ice and volatile compounds such as methane. These observations will elucidate the evolution of the lunar water cycle and delineate subsurface reservoirs of water, oxygen, and fuel necessary for sustained human presence. At the same time, lava tubes, with their natural radiation shielding and structural integrity, have emerged as favorable locations for habitat emplacement. GPR profiles will enable the generation of high-resolution subsurface models that will form the basis for rigorous site-selection criteria and resource-prospecting methodologies. Looking forward, upcoming exploration campaigns will drive GPR technology toward higher spatial resolution and deeper penetration, enabling imaging of features from meter-scale details to kilometer-scale structures. Equally important will be the adoption of multi-modal sensing methods, including extended polarization measurements, to support real-time characterization of complex geological units and volatile distributions. Finally, the fusion of GPR profiles with spectral remote-sensing data and laboratory analyses of returned regolith samples will establish a comprehensive framework for lunar resource assessment and base development. Collectively, these advances in instrumentation and data integration will deepen our understanding of lunar geology, underpin robust *in situ* resource utilization strategies and lay the scientific and technological foundation for a sustainable human presence on the Moon.

To address the demands of forthcoming lunar exploration missions, GPR technology is anticipated to advance in three principal areas: (i) lunar radar technology will advance toward higher resolution and deeper penetration capabilities. Future radar systems will surpass current depth limitations, enabling high-precision imaging of subsurface structures. These advancements will extend detection capabilities to several hundred meters, or even to kilometer-scale depths, allowing for more detailed investigations of the Moon's interior. (ii) Future lunar radar systems will integrate multi-modal sensing and expanded polarization capabilities, enhancing adaptability and efficiency in characterizing surface and subsurface materials. Real-time imaging of complex geological structures will enable more precise analyses of lunar evolution, mineral composition, and resource distribution. (iii) Progressing detection technologies will enable tighter integration between GPR data, spectral remote sensing, and laboratory analyses of returned regolith samples, providing comprehensive support for future lunar resource exploitation and base construction initiatives.

As lunar exploration progresses, advancements in GPR technology will emphasize higher resolution, multi-modal detection, deeper penetration, and enhanced polarization capabilities. Continuous innovation in radar instrumentation and the accumulation of high-quality datasets will drive lunar geological and resource research into a new era, ultimately laying the groundwork for establishing permanent human settlements on the Moon.

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Conflict of interest statement

None declared.

Data availability

Data associated with this research are available and can be obtained by contacting the corresponding author.

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